Progress Report

Lexical Structure of Advertising Interviews: Wordcloud and Token Distribution Analysis

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Abstract—
This report extends the previous lexical distance analysis by exploring the full linguistic structure of the interview corpus through token-level analysis and visualization. Using spaCy-based POS tagging, we extract adjectives, nouns and verbs from 800 simulated interviews and analyze their distributions across advertising responses for Orange and Bouygues Telecom. The notebook generates multiple wordcloud visualizations, token-frequency bar charts, lexical richness metrics and Jaccard similarity distributions. These visual diagnostics provide insight into how linguistic structure differs across brands and across interview questions.

1. OBJECTIVE OF THE ANALYSIS

This stage of the project focuses on exploring the linguistic structure of the interview corpus beyond adjective overlap. While the previous analysis quantified lexical similarity between brands, the current notebook investigates:

- Distribution of lexical categories (adjectives, nouns, verbs)
- Token frequency structures
- Brand-specific lexical signatures
- Lexical richness per interview question
- Overlap patterns between brands

The analysis relies heavily on visual exploration through wordclouds and token-frequency plots, which provide intuitive representations of lexical salience in the dataset.

2. TOKEN EXTRACTION PIPELINE

The notebook uses the **spaCy** French language model `fr\_core\_news\_lg`. Each interview response is processed through the following pipeline:

1. Load interview dataset
2. Concatenate relevant text columns
3. Apply POS tagging
4. Filter tokens by part-of-speech
5. Lemmatize tokens
6. Aggregate frequencies across the corpus

Three lexical categories are extracted:

- adjectives
- nouns
- verbs

These token lists are stored separately and used to compute frequency distributions and visualization outputs.

3. ADJECTIVE ANALYSIS

3.1. Global Adjective Wordcloud

Figure 1 shows the wordcloud constructed from all adjectives extracted from the corpus.

Figure 1. Wordcloud of all extracted adjectives.

This visualization highlights the most frequent evaluative descriptors associated with the advertisements. Words such as *fiable*, *simple*, *original* and *positif* dominate the corpus.

These terms correspond to two distinct semantic dimensions:

- reliability / clarity (often associated with Orange)
- creativity / originality (often associated with Bouygues)

3.2. Top Adjective Frequencies

The frequency distribution of the most common adjectives is shown in Figure 2.

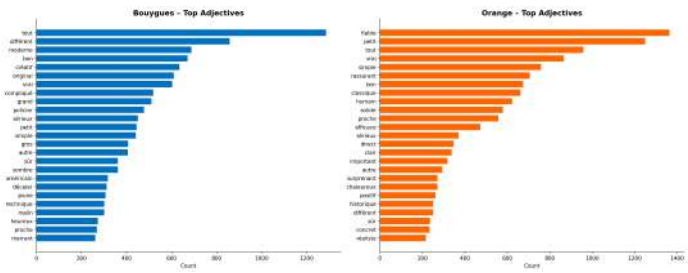

Figure 2. Top adjective frequencies across the corpus.

Unlike the wordcloud visualization, the bar chart provides exact counts and reveals the dominance hierarchy between descriptors.

4. BRAND-SPECIFIC WORDCLOUDS

To investigate how lexical patterns differ between brands, adjectives were separated according to the ad being described.

4.1. Orange Wordcloud

Figure 3. Adjective wordcloud for Orange advertisements.

The Orange wordcloud shows strong concentration around reliability and clarity-related descriptors, such as:

- fiable

Continuation of the adjective-distance analysis. This report focuses on the exploratory lexical analysis performed on the interview corpus through token extraction, wordcloud visualization, lexical richness measures and distributional comparisons across brands and questions.

- simple
- clair
- humain

This lexical cluster aligns with the brand's positioning around network reliability and stability.

4.2. Bouygues Wordcloud

Figure 4. Adjective wordcloud for Bouygues advertisements.

The Bouygues wordcloud displays a different vocabulary dominated by creativity-related terms such as:

- bizarre
- original
- créatif
- drôle

The contrast between the two clouds visually confirms the earlier quantitative results regarding lexical divergence.

4.3. Combined Comparison

The combined wordcloud visualizes the adjective vocabulary used across both brands simultaneously. Unlike the brand-specific clouds, this figure does not separate Orange and Bouygues vocabularies but instead displays the global distribution of descriptors used in the interview corpus.

Words appearing in the cloud may therefore belong to three categories:

- adjectives shared between both brands,
- adjectives predominantly associated with Orange,
- adjectives predominantly associated with Bouygues.

Shared vocabulary terms typically correspond to general evaluative descriptors common in telecom advertising, such as reliability or clarity. In contrast, brand-dominant adjectives reveal the distinct semantic framing activated by each advertisement.

This combined visualization therefore provides a global view of the lexical space before the subsequent analyses isolate brand-exclusive vocabularies and quantify lexical overlap through Jaccard similarity.

Figure 5. Comparison of adjective vocabularies between brands.

5. NOUN ANALYSIS

Adjectives capture evaluation, but nouns provide information about the conceptual objects or themes invoked by the advertisements.

5.1. Global Noun Wordcloud

Figure 6. Wordcloud of nouns extracted from the interviews.

Common nouns include references to:

- technology
- service
- network
- advertising narratives

5.2. Brand-Specific Noun Clouds

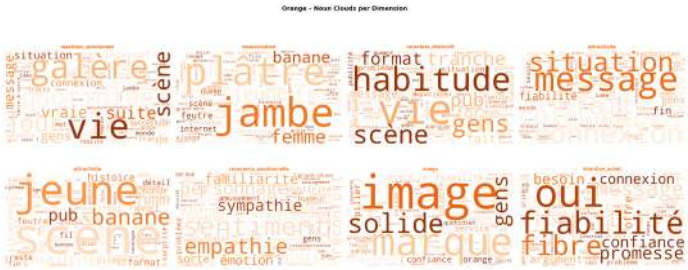

Figure 7. Noun wordcloud for Orange responses.

Figure 8. Noun wordcloud for Bouygues responses.

These figures suggest that Orange responses emphasize network infrastructure concepts, while Bouygues responses reference narrative or entertainment elements.

6. VERB ANALYSIS

Verbs capture perceived actions or processes described in the interviews.

6.1. Verb Wordcloud

Figure 9. Verb wordcloud extracted from interview responses.

The most frequent verbs include actions related to perception and communication such as:

- voir
- montrer
- expliquer
- rappeler

6.2. Brand-Specific Verb Clouds

Figure 10. Verb cloud for Orange responses.

Figure 11. Verb cloud for Bouygues responses.

7. LEXICAL OVERLAP ANALYSIS

In order to quantify how similar the vocabularies used for describing the two advertisements are, lexical overlap is measured using the **Jaccard similarity coefficient**. This metric is widely used in information retrieval and text mining to compare the similarity between two sets of tokens.

For two sets of words A and B, the Jaccard similarity is defined as:

$$J(A,B) = \frac{|A \cap B|}{|A \cup B|} \tag{1}$$

where:

- A represents the set of tokens used by a respondent when describing the **Bouygues** advertisement,
- B represents the set of tokens used when describing the **Orange** advertisement,
- $|A \cap B|$ is the number of tokens shared between both sets,
- $|A \cup B|$ is the total number of unique tokens used across both sets.

The Jaccard coefficient ranges between 0 and 1:

- $J = 0$ indicates **no lexical overlap** between the two sets,
- $J = 1$ indicates **identical vocabularies**.

In this analysis, Jaccard similarity is computed at the **panelist level**. For each interview respondent, two sets of tokens are constructed from their answers describing the two advertisements. The similarity score therefore captures how much the same respondent uses the same vocabulary to describe both brands.

This approach allows us to measure lexical similarity independently of overall corpus size or token frequency, focusing purely on vocabulary overlap.

7.1. Adjective Jaccard Distribution

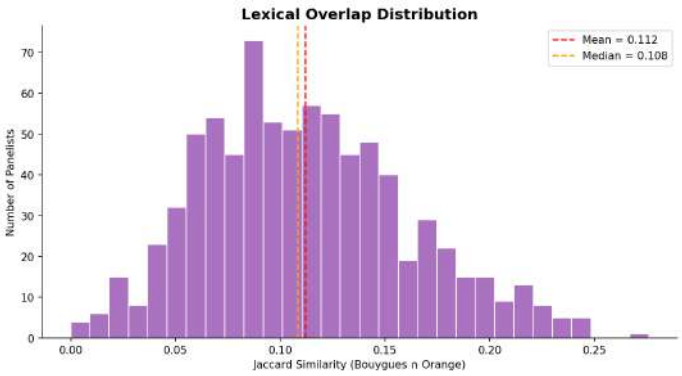

Figure 12. Distribution of adjective Jaccard similarity across panelists.

Adjectives primarily encode *evaluative judgments* and *emotional reactions*. The distribution in Figure X shows that the majority of respondents exhibit relatively low Jaccard similarity values.

This indicates that respondents tend to use **distinct evaluative vocabularies** when describing the two advertisements. In other words, the semantic fields used to characterize the brands differ substantially at the level of perceived qualities.

The presence of a concentration of values close to zero suggests that many respondents use entirely different descriptors for the two advertisements, reinforcing the idea that the campaigns activate different evaluative frames.

7.2. Noun Jaccard Distribution

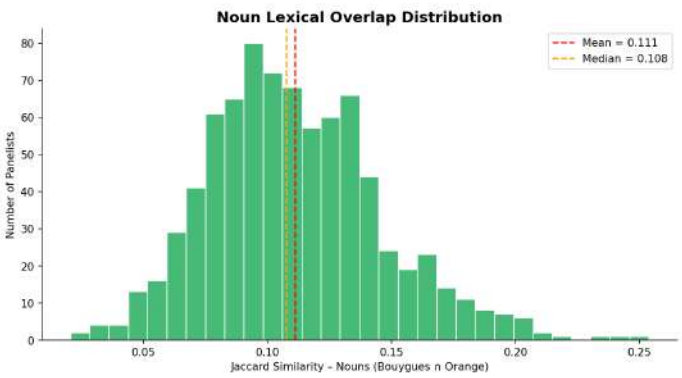

Figure 13. Distribution of noun Jaccard similarity.

Nouns capture the **conceptual objects, actors, and themes** referenced in the advertisements. Compared to adjectives, the noun Jaccard distribution tends to be more centered and slightly higher.

This reflects the fact that respondents frequently refer to similar structural elements across both advertisements, such as the brand, technological infrastructure, or narrative components of the advertisement. Since both advertisements belong to the same telecom domain, some conceptual vocabulary is naturally shared.

Nevertheless, the distribution still shows considerable variation across panelists, indicating that the two advertisements evoke different conceptual references in many responses.

7.3. Verb Jaccard Distribution

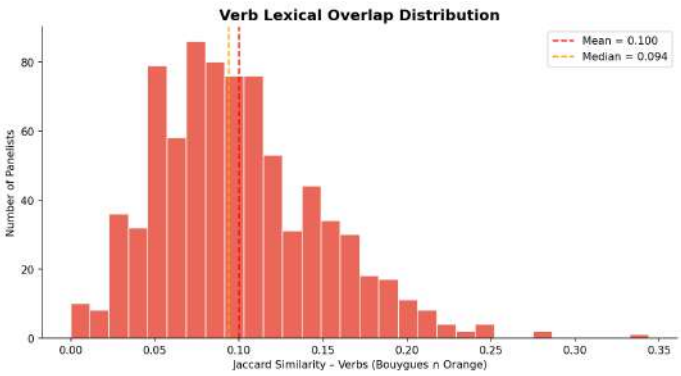

Figure 14. Distribution of verb Jaccard similarity.

Verbs describe **actions, processes, and cognitive interpretations**. In interview responses, verbs often correspond to how respondents describe the narrative or message of the advertisement (e.g., showing, explaining, illustrating).

The verb similarity distribution tends to lie between the noun and adjective distributions. This suggests that while respondents often describe similar types of actions (for example, explaining a service or showing a scenario), the way these actions are framed still varies between the two advertisements.

7.4. Interpretation

Taken together, the three distributions reveal a clear hierarchy in lexical overlap across linguistic categories:

- **Adjectives** show the **lowest overlap**, indicating that evaluative judgments about the advertisements differ strongly between brands.
- **Nouns** show higher overlap, reflecting shared conceptual elements of telecom advertising.
- **Verbs** occupy an intermediate position, capturing partially shared narrative structures.

This pattern is consistent with expectations from advertising analysis. While the two campaigns belong to the same product category and therefore share some conceptual vocabulary, they appear to evoke distinct emotional and evaluative responses among respondents.

The low adjective similarity in particular suggests that each advertisement activates a different **semantic framing** of the brand in the minds of viewers. Such differences in evaluative language are especially important in marketing contexts, as they reflect the perceptual positioning of competing brands.

8. LEXICAL RICHNESS ACROSS QUESTIONS

Beyond simple frequency analysis, the notebook also investigates *lexical richness* across the different interview questions. Lexical richness measures how diverse the vocabulary is within responses. In this context, it is defined as the number of **unique lemmatized tokens** of a given part-of-speech (POS) category used when answering a question.

Formally, for a given question q, lexical richness is computed as:

$$R_q = |\{w : w \in V_q\}| \tag{2}$$

where V_q denotes the set of unique lemmatized tokens extracted from all responses to question q. Higher values indicate a greater diversity of vocabulary used by respondents.

This measure provides insight into **how cognitively open or constrained a question is**. Questions prompting emotional or narrative reflection tend to generate richer lexical variation, while factual or binary questions tend to produce more repetitive vocabulary.

The analysis is conducted separately for three linguistic categories: adjectives, nouns and verbs. Each category captures a different dimension of how respondents linguistically construct their perception of the advertisements.

8.1. Adjective Richness

Adjectives primarily encode **evaluative and affective judgments**. In advertising analysis, they often describe perceived qualities of the brand or advertisement, such as credibility, creativity, clarity, or emotional tone.

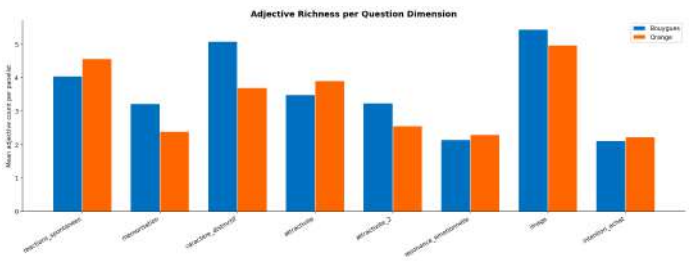

Figure 15. Adjective richness per interview question.

The adjective richness plot shows that certain questions elicit a much broader evaluative vocabulary than others. In particular, questions that ask respondents to describe their *emotional reaction*, *overall impression*, or *memorability* tend to produce a higher number of unique adjectives.

This suggests that these questions encourage respondents to articulate nuanced subjective judgments, leading to a wider semantic range of descriptors. In contrast, questions focusing on factual recall or preference selection produce lower adjective diversity, as respondents tend to reuse common evaluative terms.

8.2. Noun Richness

Nouns capture the **conceptual objects and themes** referenced by respondents. In the context of advertising interviews, these typically correspond to elements such as:

- narrative elements in the advertisement
- technological infrastructure (e.g., network, service)
- characters or scenarios depicted in the ad
- brand-related concepts

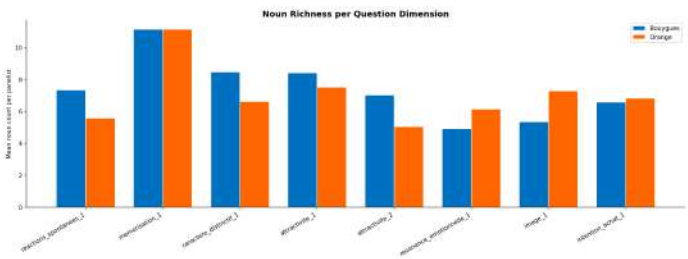

Figure 16. Noun richness per interview question.

Compared to adjectives, noun richness tends to be more stable across questions. This reflects the fact that respondents repeatedly refer to a relatively limited set of conceptual elements present in the advertisements themselves.

However, questions asking respondents to *describe what happens in the advertisement* or *recall specific elements* typically generate higher noun richness. These questions prompt respondents to mention different objects, scenes, or actors appearing in the ads.

8.3. Verb Richness

Verbs encode the **actions and processes** perceived in the advertisements or described by respondents. In interview responses, verbs often represent cognitive processes (e.g., *penser*, *comprendre*), perceptual actions (e.g., *voir*, *montrer*), or narrative events occurring within the advertisement.

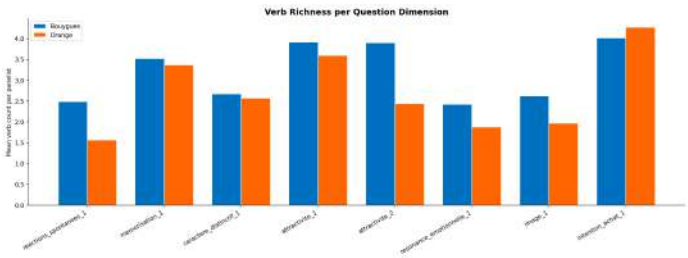

Figure 17. Verb richness per interview question.

Verb richness tends to be lower than noun richness but higher than adjective richness in some contexts. This reflects the structured nature of how respondents describe advertisements: many responses follow a similar narrative structure (e.g., describing what happens in the ad or how the message is communicated).

Questions prompting respondents to *explain the message of the advertisement* or to *describe what happens in the scene* tend to generate the highest verb diversity, as they require respondents to reconstruct the sequence of actions portrayed in the advertisement.

8.4. Interpretation

Taken together, the three richness measures provide insight into how different interview questions activate different linguistic dimensions of consumer perception:

- **Adjectives** capture evaluative and emotional judgments about the advertisement.
- **Nouns** represent the conceptual objects and themes referenced by respondents.
- **Verbs** describe actions, narrative events and cognitive processes.

The richness patterns suggest that open-ended questions about *emotional reactions*, *overall impressions* and *memorability* generate the most linguistically diverse responses. In contrast, questions focused on specific recall or preference tend to produce more constrained vocabularies.

This observation is consistent with expectations from survey design: questions that invite interpretative or reflective responses naturally produce greater lexical diversity than those requiring direct evaluation or simple comparison.

9. EXCLUSIVE VOCABULARY

Finally, the notebook isolates tokens that appear exclusively in responses to one brand.

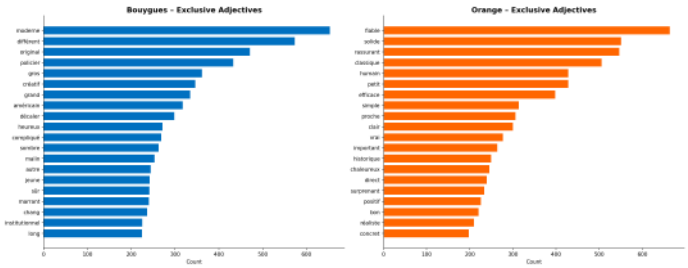

Figure 18. Adjectives appearing only for one brand.

Figure 19. Adjectives appearing only for one brand.

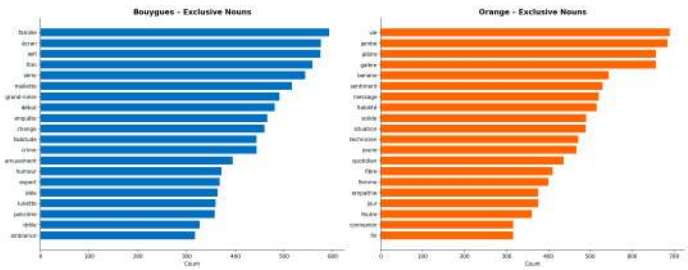

Figure 20. Brand-exclusive nouns.

Figure 21. Brand-exclusive nouns.

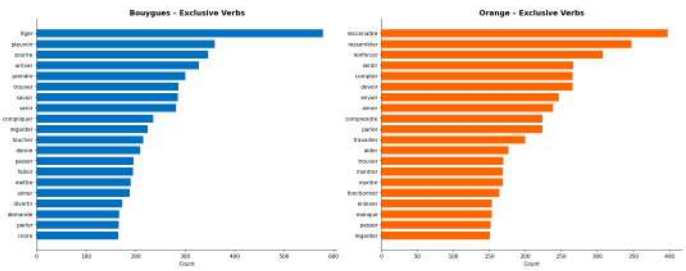

Figure 22. Brand-exclusive verbs.

Figure 23. Brand-exclusive verbs.

These figures illustrate the asymmetric vocabulary structure between brands, which helps explain the divergence observed in the earlier distance metrics.

10. SUMMARY

This notebook provides a detailed exploratory analysis of the lexical structure of the interview corpus. Through POS-tagging, token frequency analysis and visualization techniques such as wordclouds and Jaccard similarity distributions, we observe clear linguistic differences in how respondents describe the two advertisements.

Adjectives appear to be the most brand-specific lexical category, while nouns and verbs show higher overlap across responses. Wordcloud visualizations provide an intuitive representation of these differences and confirm the semantic clusters suggested by earlier statistical analysis.